

MERSING, Malaysia

WMO: 486740

Lat: 2.4450N

Lon: 103.8310E

Elev: 148

StdP: 14.62

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	71.6	73.0	69.5	108.9	76.3	69.9	110.4	76.1	18.9	80.4	17.5	80.6	5.6	240	0.343

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	15.8	93.5	83.3	92.8	83.0	91.5	82.4	84.7	92.0	84.1	91.2	83.3	90.2	7.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.7	171.3	91.0	82.2	168.8	90.6	80.9	161.6	89.1	49.1	92.4	48.4	90.6	47.3	91.3	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.8	14.0	12.5	70.5	95.6	1.3	2.3	69.6	97.2	68.8	98.5	68.0	99.8	67.1	101.5		
(5)			69.3	86.6	1.2	1.3	68.5	87.6	67.8	88.3	67.1	89.0	66.3	89.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.7	80.3	81.2	81.9	82.7	83.3	82.8	82.2	82.0	81.8	81.7	80.9	80.1
	DBStd	2.12	2.35	2.00	2.16	1.93	1.88	1.89	1.76	1.62	1.55	1.66	1.76	2.15
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11587	940	874	988	980	1032	984	999	992	955	983	926	934
	CDD65	6112	475	454	523	530	567	534	534	527	505	518	476	469
	CDH74	58123	4341	4524	5473	5560	5761	5213	4941	4784	4496	4670	4185	4175
	CDH80	20599	967	1207	1935	2282	2447	2128	1921	1846	1753	1788	1323	1002
Wind	WSAvg	6.2	8.8	8.7	6.7	5.2	5.1	5.4	5.6	5.9	5.5	5.0	5.1	7.3
Precipitation	PrecAvg	103.3	12.6	5.0	5.2	4.8	5.7	5.7	6.4	6.8	6.5	8.0	13.8	23.9
	PrecMax	144.6	34.6	19.6	14.2	13.0	12.9	11.3	12.2	12.7	13.4	16.9	25.2	60.5
	PrecMin	65.8	0.2	0.0	0.0	0.2	1.9	0.9	0.0	2.4	0.1	2.0	1.3	2.9
	PrecStd	21.1	9.1	5.1	3.5	3.1	2.5	2.1	2.4	2.2	3.1	3.5	5.9	12.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.9	89.8	91.5	95.1	96.4	94.9	93.5	93.6	93.0	93.2	91.4	89.9
		MCWB	81.1	80.6	82.3	85.2	84.5	82.2	83.2	81.6	82.9	82.8	82.1	80.7
	2%	DB	87.8	88.1	89.9	92.9	93.5	93.0	91.7	91.5	91.5	91.4	89.6	87.9
		MCWB	80.4	79.9	80.8	83.7	83.9	82.8	82.4	82.0	82.4	82.1	80.9	80.3
	5%	DB	86.0	87.4	89.2	91.2	91.7	91.4	90.0	89.9	89.9	89.8	88.0	86.2
		MCWB	79.0	79.5	80.4	82.3	82.9	82.4	81.7	81.6	81.5	81.3	80.6	79.3
	10%	DB	84.4	85.9	87.8	89.6	90.0	89.7	89.3	88.2	88.2	88.1	86.4	84.5
		MCWB	77.9	78.6	79.9	81.4	82.1	81.6	81.5	81.1	81.0	81.0	79.7	78.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.2	81.5	83.3	85.5	85.7	85.5	84.9	84.4	84.4	84.2	82.8	82.3
		MCDB	87.7	88.1	89.8	93.8	93.5	92.3	91.7	90.7	91.0	91.1	89.2	87.9
	2%	WB	80.7	80.2	81.7	84.3	84.6	84.2	83.7	83.1	83.3	82.8	81.7	80.7
		MCDB	86.3	86.5	88.6	91.8	92.3	91.2	90.4	89.6	90.1	89.7	87.8	86.1
	5%	WB	79.4	79.3	80.7	82.9	83.7	83.0	82.5	82.3	82.4	82.1	80.8	79.7
		MCDB	84.2	85.2	87.5	89.9	91.2	89.7	89.0	88.8	89.2	88.8	86.8	84.6
	10%	WB	78.7	78.8	79.8	81.9	82.7	82.0	81.5	81.3	81.3	81.0	80.0	79.0
		MCDB	83.0	84.4	86.0	88.5	89.6	88.4	87.8	87.6	87.8	87.4	85.5	83.5
Mean Daily Temperature Range	5% DB	MDBR	8.0	9.3	11.7	14.6	15.8	15.3	15.0	15.1	15.4	14.7	12.2	9.2
		MCDBR	10.4	11.1	13.5	16.5	17.4	16.9	16.4	16.4	16.6	16.4	14.4	11.7
		MCWBR	5.6	5.4	6.7	8.6	9.2	8.9	9.1	9.2	9.3	9.0	7.6	6.2
	5% WB	MCDBR	9.8	10.3	12.7	16.0	16.7	16.2	15.9	15.7	16.2	15.8	13.6	11.0
		MCWBR	5.6	5.4	6.8	9.2	9.6	9.5	9.5	9.6	9.7	9.3	7.5	6.2
Clear-Sky Solar Irradiance	taub	0.461	0.460	0.464	0.464	0.479	0.501	0.491	0.526	0.547	0.557	0.457	0.463	
	taud	2.321	2.309	2.301	2.299	2.262	2.196	2.236	2.118	2.057	2.013	2.334	2.325	
	Ebn at Noon	275	277	275	269	258	248	253	249	249	248	274	272	
	Edh at Noon	42	43	44	43	43	45	44	51	55	58	41	42	
All-Sky Solar Radiation	RadAvg	1373	1643	1680	1692	1590	1496	1495	1528	1518	1505	1348	1257	
	RadStd	124	149	141	85	69	82	93	63	88	106	78	121	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	+0.59	+0.18	N/A	+0.43	+0.63	+0.58	N/A	N/A	+162	+158

Nomenclature: See separate page